

GENERIC REFINERY - GULF COAST

100 kBPSD Crude & Vacuum Distillation Unit

Control Philosophy & Proposed Control Scheme

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1. Introduction and scope

This document defines the control philosophy and the proposed control scheme for the 100 kBPSD Crude & Vacuum Distillation Unit (design throughput 100,000 BPSD of Arab Light (design), turndown 50 %). It covers the regulatory (BPCS / DCS) layer, the advanced process control (APC / MPC) layer, alarm management, the safety layer interfaces (SIS, BMS, F&G - detailed in CFU-000-IC-RPT-002 and CFU-000-IC-CE-001), the ICS architecture and the instrument design criteria, including control valve sizing for the key valves.

All tags follow ISA-5.1 and are taken from the master loop list `cfu/control_loops.py` (data/control_loops.json) and the instrument index CFU-000-IC-IDX-001 (data/instruments.json). Tags introduced by I&C (analysers, computing blocks) are listed in section 13 for inclusion in the index.

Related deliverables

Document	Title
CFU-100-IC-CSD-001..005, CFU-000-IC-CSD-006	Proposed control scheme diagrams (A1)
CFU-000-IC-BLK-001	ICS architecture block diagram (A1)
CFU-000-IC-CE-001	Cause & effect matrix (xlsx + A1 drawing)
CFU-000-IC-RPT-002	SIF list and SIL determination (LOPA)
CFU-000-IC-IOL-001	I/O list (xlsx), data/io_list.json
CFU-000-IC-CAL-001	Control valve sizing (xlsx, summarised in section 12)
CFU-100-IC-LD-001..003	Typical loop diagrams (A3)
CFU-xxx-PR-PFD / PID	Process flow diagrams / P&IDs (process discipline)

Codes and standards: ISA-5.1, ISA-5.4, ISA-18.2 / IEC 62682, EEMUA 191, ISA-75.01.01 / IEC 60534, IEC 61511 / ISA-84, IEC 62443, API 551, API 552, API 554, API 556, NFPA 85/86, IEC 60079.

2. Control objectives

- **Safety and environment** - keep the unit within its safe operating envelope; independent protection layers (SIS / BMS / F&G) separate from the BPCS; no single failure leads to an unsafe state.
- **Product quality** - hold naphtha end point, kerosene flash / freeze, diesel T95 / cloud, AGO and VGO quality and LPG / naphtha RVP within specification with minimum give-away.
- **Throughput** - maximise crude charge up to the active constraint (heater firing, flooding, overhead condenser, vacuum system, pump / valve limits).
- **Energy** - maximise preheat recovery (CIT), minimise excess O2, stripping steam and reflux.
- **Stability and operability** - smooth feed to downstream units (averaging level control), fast crude switches, robust start-up / shutdown and 50 % turndown.
- **Reliability** - redundant controllers, I/O isolation per area, dual-fed power, no common-mode with the SIS.

The control hierarchy is: L0/L1 field and regulatory PID (DCS), L2 operator HMI and supervisory logic, L3 multivariable predictive control with inferentials and an economic LP; the SIS / BMS / F&G act independently of all of these.

Key operating values (design case, data/process_results.json)

Item	Value	Item	Value
Crude charge	662 m3/h std (570 t/h)	Desalter temperature	136 °C
CIT (H-101 inlet)	274 °C	H-101 COT	366 °C
H-101 absorbed / fired	60.7 / 67.4 MW	H-101 passes / burners	8 / 16
C-101 top T / P	134 °C / 1.19 barg	D-102 pressure	0.69 barg
Flash zone T / P	361 °C / 1.47 barg	Reflux	101 t/h
TPA / MPA / BPA duty	8.1 / 12.8 / 12.4 MW	Kero / diesel / AGO draw T	220 / 302 / 343 °C
H-201 COT	398 °C	C-201 top / flash zone	20 / 60 mbar(a)
LVGO / HVGO PA duty	9.0 / 13.6 MW	C-201 bottoms T	366 °C (limit 365 °C)

Item	Value	Item	Value
C-105 top P	11.8 bar(a)	C-106 top P	2.1 bar(a)

3. Regulatory control strategy

General rules applied to all sections:

- Flow loops are the innermost loops for all manipulated streams (cascade slaves); valves are never positioned directly by quality or temperature controllers.
- Product draws from towers are on flow control; accumulator / bottoms levels cascade to the downstream flow (material balance 'in the direction of flow' except where noted).
- Pumparound duties are controlled by return temperature via exchanger bypass on the pumparound side, with circulation rate on flow control.
- Ratio stations (FFIC / FFY) relate chemical injection, wash water and stripping steam to their master flows; ratios are operator / MPC set.
- Selectors and cross-limits are implemented in the DCS with anti-windup (external reset feedback).
- All cascades are bumpless; on slave failure or bad PV the master goes to tracking.

3.1 Crude charge and desalting (CFU-100-IC-CSD-003)

- **Throughput** - FIC-1001 on P-101 discharge (FV-1001) is the unit throughput master (662 m³/h design). Its SP is set by the operator or the CDU MPC (max-feed push). The PV / SP is broadcast on a ratio / feed-forward bus to: demulsifier FFIC-1002 (X-101 stroke), wash water FFIC-1003 (5 vol %), caustic FFIC-1010 (X-102), the H-101 pass flow controllers and the COT feed-forward.
- **Desalter temperature** - TIC-1004 positions TV-1004 on the crude-side bypass of E-105 (VR side always flowing) to hold 136 °C (limits 125-145 °C).
- **Mixing** - PDIC-1005 / 1006 hold the mix-valve dP (0.5-1.5 bar) for wash-water dispersion; the SP is optimised against salt-in-crude AT-1046 and oil-in-brine AT-1049 (operator, not closed loop).
- **Interface** - LIC-1007 / 1008 (guided-wave radar + density profiler) control the water/oil interface tightly via brine valves LV-1007 / LV-1008; second-stage brine is recycled counter-currently to stage 1. Low-low interface (SIF-107) trips the transformers.
- **Pressure** - PIC-1009 holds the desalter outlet pressure (10.0 barg) above crude vapour pressure at the booster pump P-102 suction.

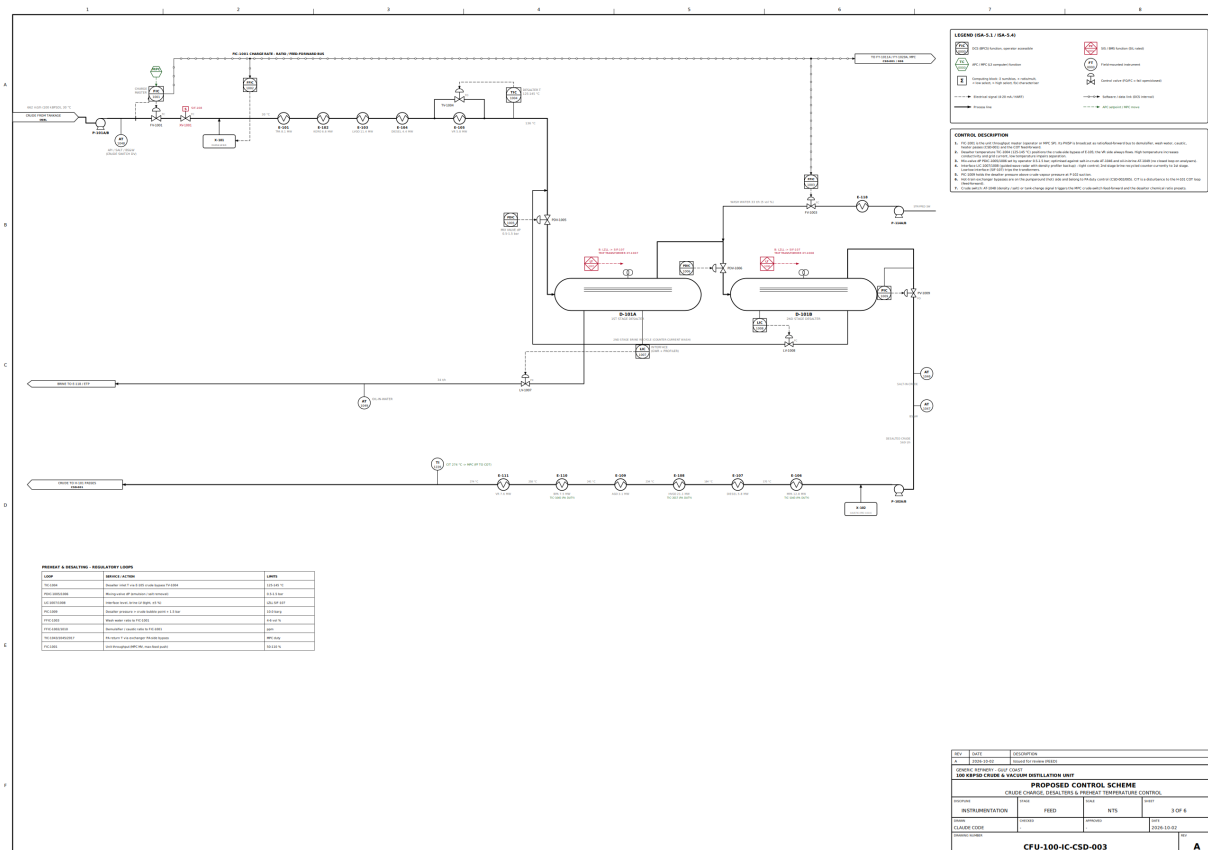


Figure 1 - Crude charge, desalting and preheat control (CFU-100-IC-CSD-003)

3.2 Preheat train

The cold train (E-101..E-105) heats crude from 30 °C to the desalter; the hot train (E-106..E-111) to a CIT of 274 °C. Exchanger bypasses that carry a control function are on the hot (pumparound) side and belong to the pumparound duty loops: TIC-1043 (MPA, E-106), TIC-1045 (BPA, E-110 / E-113) and TIC-2017 (HVGO, E-108). This keeps the crude flow path simple and makes CIT a measured disturbance (TI-1226) fed forward to the COT controller. The MPC trades heat recovery (higher CIT) against column fractionation through the PA duties.

3.3 Atmospheric heater H-101 (CFU-100-IC-CSD-001)

- Pass flow** - each of the 8 passes has a flow controller FIC-1011..1018. The base SP of each pass = FIC-1001 / 8 (FY-1011A, 71.1 t/h).
- Pass balancing** - TDIC-1019 compares the pass outlet temperatures TI-1011..1018 with their average and computes biases FY-1011B..1018B; the biases are normalised so that their sum is zero, i.e. **total flow is held constant** and only the distribution changes. Biases are clamped to $\pm 10\%$ of pass flow and frozen when any pass FIC is not in cascade or a pass flow is near its low-low trip.
- COT** - TIC-1020 (SP from MPC) is the master. A feed-forward FY-1020A = $f(\text{charge} \times (\text{COT} - \text{CIT}))$ with lead-lag dynamic compensation is added (FY-1020B) to give the firing demand.
- Cross-limiting (lead-lag) combustion control** - fuel SP = MIN(firing demand, air available / stoichiometric ratio) (FY-1021A); air SP = MAX(firing demand, actual fuel heat release) $\times$ air/fuel ratio $\times$ O₂ trim (FY-1025A / B). On a load increase the air leads and fuel follows; on a decrease the fuel leads and air follows, so the firebox never becomes sub-stoichiometric. Actual fuel heat release uses FG flow FT-1021 corrected by the Wobbe analyser AT-1021.
- Fuel** - the fuel heat demand is characterised to a burner pressure SP (burner curve, FY / PY-1021A); a min-fire stop (high select PY-1021B) keeps burners above the stable minimum. PIC-1021 manipulates PV-1021. (The loop list describes this as 'low select vs min-fire'; the implementation is a high select against the minimum burner pressure combined with the cross-limit low select.)

- **O₂ trim** - AIC-1022 (arch O₂, typical SP 2-3 % wet, with CO override > 200 ppm) multiplies the air/fuel ratio, limited ± 10 %. Combustion air FIC-1025 positions the FD fan inlet vanes (K-101A/B).
- **Draft** - PIC-1023 holds -2.5 mmH₂O at the arch; split range via PY-1023: ID fan speed (K-102A/B VSD) 0-50 %, stack damper 50-100 %. On ID fan trip the damper opens for natural-draft operation at reduced firing.
- **Stripping steam superheat** - TIC-1024 (350 °C) on the convection steam coil desuperheater.

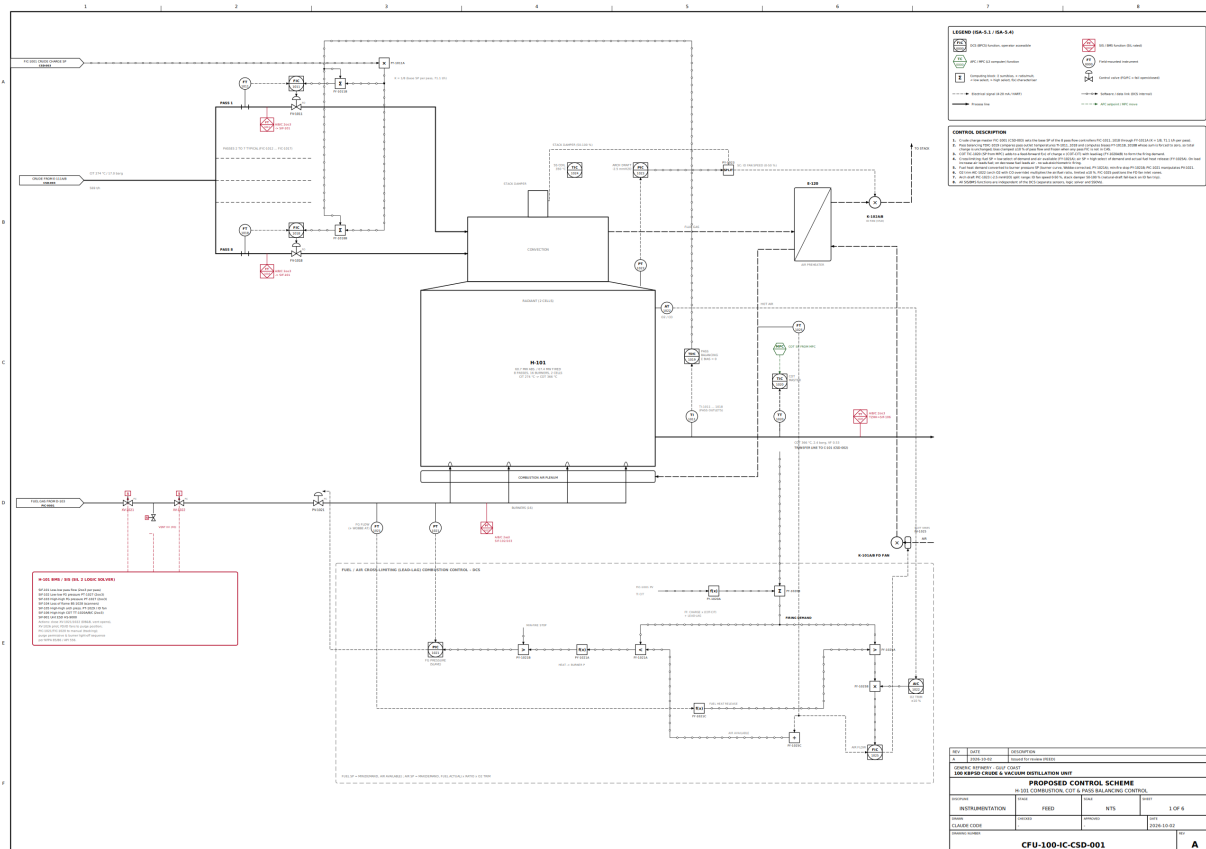


Figure 2 - H-101 combustion, COT and pass balancing (CFU-100-IC-CSD-001)

3.4 Atmospheric column C-101 and side strippers (CFU-100-IC-CSD-002)

- **Column pressure (split range)** - PIC-1032 on D-102 (0.69 barg): 0-50 % output closes the fuel-gas make-up PV-1032B, 50-100 % opens the off-gas valve PV-1032A to FG / flare. Normally the overhead is a total condenser and neither valve passes significant flow. The MPC may lower the PIC SP to improve lift, constrained by A-101 duty and the reflux drum temperature.
- **Top temperature -> reflux cascade** - TIC-1030 (top 134 °C, pressure-compensated) sets the SP of reflux FIC-1031 (101 t/h). TIC-1030 is the naphtha end-point handle for the MPC (CV: AT-1038 / inferential).
- **Overhead accumulator** - LIC-1033 (averaging) -> naphtha FIC-1034 (feed to C-105); boot interface LIC-1035 tight to LV-1035; pH AIC-1037 trims the neutraliser ratio FFIC-1036 (X-103).
- **Pumparound duty** - FIC-1040 / 1042 / 1044 hold circulation; TIC-1041 / 1043 / 1045 set the return temperature, i.e. the duty, via PA-side exchanger bypass (TV-1041, TV-1043, TV-1045). Duty distribution TPA / MPA / BPA is an MPC degree of freedom (heat recovery vs. internal reflux in the fractionation zones).
- **Side-draw cut-point control** - draw flows FIC-1050 / 1060 / 1070 are the cut-point handles; their SPs come from the MPC (kero flash / freeze, diesel T95, AGO T95). The internal reflux below each draw must stay above a minimum (inferred from the column heat balance - MPC constraint).
- **Strippers** - bottoms level LIC-1051 / 1061 / 1071 cascades to product flow FIC-1052 / 1062 / 1072. Stripping steam FIC-1054 / 1064 / 1074 is ratioed to the product flow (FFY, lb/bbl) - the MPC adjusts the ratio for flash point.
- **Bottoms** - LIC-1082 cascades to the atmospheric residue flow FIC-1083 (also the H-201 charge). Bottom stripping steam FIC-1081 is ratioed to AR (7638 kg/h design, ~10 lb/bbl).

- **Overflash** - FI-1080 (wash-zone liquid) is monitored as % of charge with a low alarm; it is a hard MPC constraint (≥ 3 vol %) raised by COT or by reducing BPA duty. Section dP PDI-1237 is the flooding indicator.

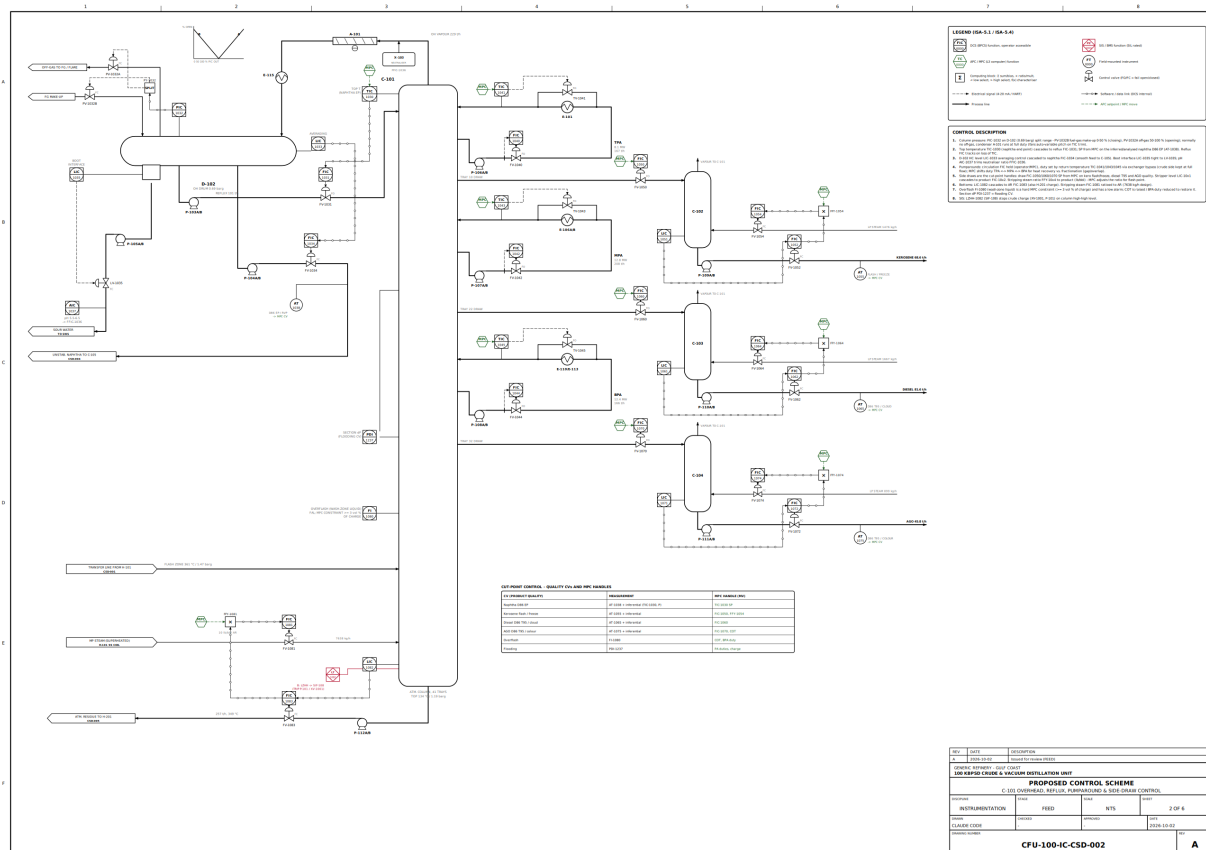


Figure 3 - C-101 overhead, reflux, pumparound and side-draw control (CFU-100-IC-CSD-002)

3.5 Stabiliser C-105 and splitter C-106 (CFU-100-IC-CSD-004)

- **Pressure** - C-105: PIC-1091 hot-vapour bypass PV-1091 around the flooded condenser A-106 (11.8 bar(a)). C-106: PIC-1100 on the flooded-condenser outlet PV-1100 (2.1 bar(a)); floating-pressure operation under MPC.
- **Material balance** - D-105 / D-106 levels LIC-1092 / 1101 cascade to LPG FIC-1093 / LN FIC-1102; C-105 bottoms LIC-1097 to LV-1097 (C-106 feed, averaging); C-106 bottoms LIC-1106 to HN FIC-1107.
- **Energy balance** - sensitive-tray temperatures TIC-1095 (C-105) and TIC-1104 (C-106, tray 30), pressure-compensated, cascade to reboiler steam FIC-1096 (HP steam, E-116) and FIC-1105 (MP steam, E-117). Reflux: FIC-1094 on ratio to feed (FFY-1094) for C-105; FIC-1103 on flow for C-106.
- **Dual-composition strategy** - top composition (LPG C5+, LN C6+) is controlled with reflux (L/F), bottom composition (naphtha RVP, HN C5- / IBP) with the sensitive-tray temperature SP. The interaction (L-V configuration, RGA ~ 2-4) is handled by the light-ends MPC using GC analysers AT-1099, AT-1108 and AT-1109. Without MPC the columns run single-ended (tray temperature) with reflux on ratio and operator trim.

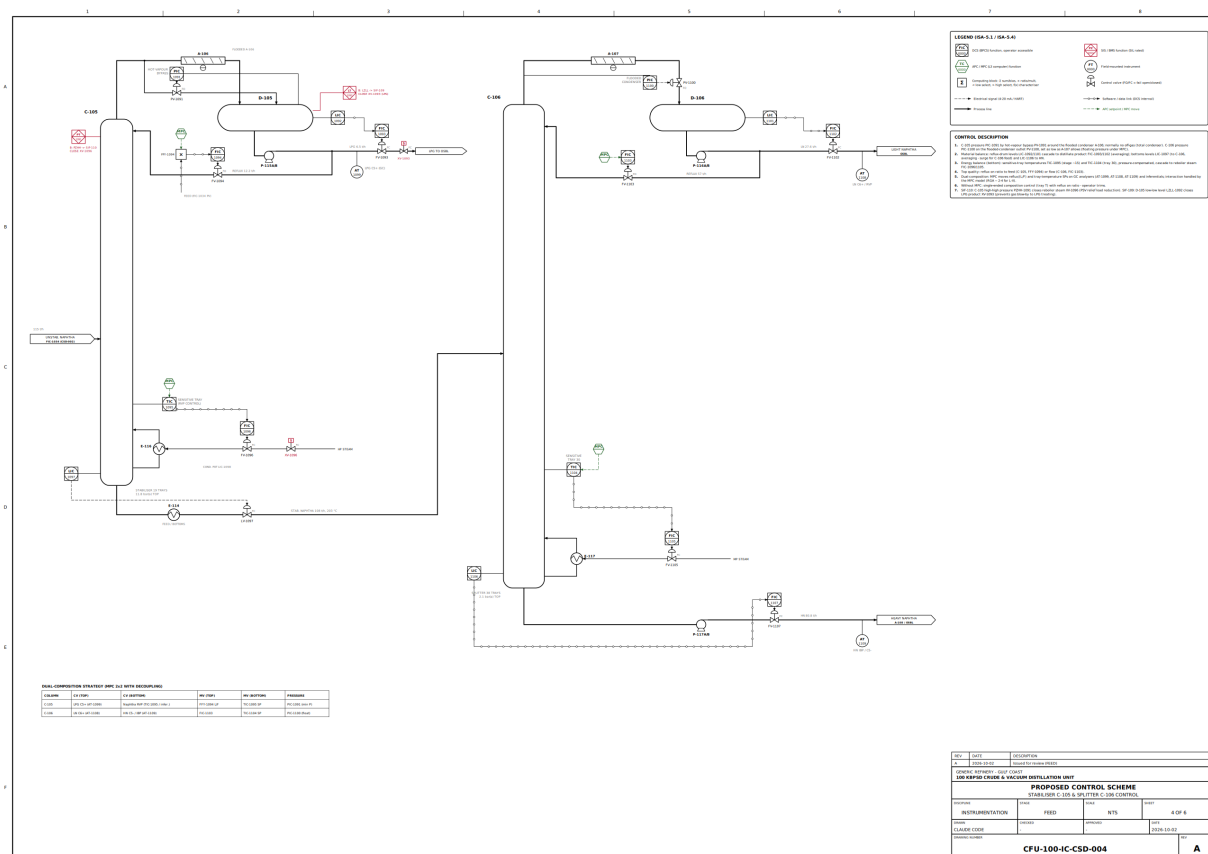
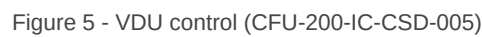


Figure 4 - Stabiliser / splitter control (CFU-100-IC-CSD-004)

3.6 Vacuum unit H-201 / C-201 (CFU-200-IC-CSD-005)

- **H-201** - pass flows FIC-2001..2004 (ratio of FIC-1083), coil (velocity) steam FIC-2007, COT TIC-2005 (398 °C) cascaded to FG pressure PIC-2006 with the same cross-limiting and FF philosophy as H-101; O2 AIC-2008 on the stack damper (natural draft).
- **Vacuum pressure** - PIC-2010 (20 mbar(a)) recycles off-gas / steam from the after-condenser to the 1st-stage ejector J-201 suction (PV-2010). Motive steam is not throttled. The MPC pushes the pressure SP to the lowest achievable value (max lift), constrained by PV-2010 output.
- **LVGO section** - top temperature TIC-2012 cascades to the LVGO PA return temperature TIC-2013 (A-201 fan pitch / bypass); PA circulation FIC-2011; pan level LIC-2014 -> product FIC-2015.
- **HVGO section** - FIC-2016 circulation, TIC-2017 PA return temperature by E-108 PA-side bypass; pan level LIC-2018 -> HVGO product FIC-2019.
- **Wash oil (minimum flow)** - FIC-2020 SP = high select (FY-2020) of the MPC SP and a minimum wetting rate f(charge), protecting the wash bed against coking. Low-flow alarm is high priority.
- **Bottoms and quench** - LIC-2024 -> VR FIC-2025; boot temperature TIC-2026 (max 365 °C) cascades to the cooled-VR quench FIC-2027. Stripping steam FIC-2023 is ratioed to charge. Slop wax LIC-2021 -> FIC-2022.



Fuel-gas header pressure PIC-9001 (D-103) with KO drum level LIC-9002; flare KO drum D-104 level LIC-9003 (pump-out P-119A/B start/stop); LP steam header PIC-9004 (let-down). Steam generators E-113 / E-201 use three-element level control (LIC-1110 / 2030: level + steam flow FF + BFW flow) with steam pressure PIC-1111 / let-down.

4. Level control tuning philosophy

Levels are classified as **averaging** (use the vessel hold-up to filter flow disturbances to the downstream unit - level allowed to swing between 25 % and 75 %) or **tight** (level itself matters: interfaces, draw pans, steam drums, seals). Surge time is the volume between 25 % and 75 % of span divided by the outflow. Averaging PI tuning: Kc 0.5-1.0 %/% and integral time ~ 4 x surge time (critically damped response to a step in inflow, level stays within the band); tight control: Kc 2-4 %/%, short integral. Error-squared or gap action is not used (non-linear behaviour hinders MPC identification).

Loop	Service	Vol. 25-75 % m3	Outflow m3/h	Surge min	Mode	Manipulates	Tuning guide
LIC-1033	D-102 HC level	66.4	165	24.2	Averaging	FIC-1034 (C-105 feed)	PI: Kc 0.5-1.0 %/%, Ti ~ 97 min
LIC-1092	D-105 level	7.6	12	37.1	Averaging	FIC-1093 (LPG)	PI: Kc 0.5-1.0 %/%, Ti ~ 148 min
LIC-1101	D-106 level	29.6	41	43.3	Averaging	FIC-1102 (LN)	PI: Kc 0.5-1.0 %/%, Ti ~ 173 min
LIC-1082	C-101 bottom (2.0 m span)	19.2	365	3.2	Averaging	FIC-1083 (H-201 charge)	PI: Kc 0.5-1.0 %/%, Ti ~ 13 min
LIC-1051	C-102 bottom (1.5 m span)	3.4	111	1.8	Averaging	FIC-1052	PI: Kc 0.5-1.0 %/%, Ti ~ 7 min
LIC-1061	C-103 bottom (1.5 m span)	4.3	131	1.9	Averaging	FIC-1062	PI: Kc 0.5-1.0 %/%, Ti ~ 8 min
LIC-1071	C-104 bottom (1.5 m span)	2.3	74	1.9	Averaging	FIC-1072	PI: Kc 0.5-1.0 %/%, Ti ~ 8 min
LIC-1097	C-105 bottom (1.5 m span)	5.2	196	1.6	Averaging	LV-1097 (C-106 feed)	PI: Kc 0.5-1.0 %/%, Ti ~ 6 min
LIC-1106	C-106 bottom (1.5 m span)	11.3	140	4.9	Averaging	FIC-1107 (HN)	PI: Kc 0.5-1.0 %/%, Ti ~ 19 min
LIC-2024	C-201 boot (2.0 m span)	14.1	138	6.1	Tight-ish (residence time limit, coking)	FIC-2025 (VR)	PI: Kc 1.5-2, Ti 5-8 min; TIC-2026 quench limits T
LIC-1007 / 1008	Desalter interface	-	36	-	Tight	LV-1007 / LV-1008	PI: Kc 2-3, Ti 3-5 min (grid protection)
LIC-1035	D-102 boot interface	-	13	-	Tight	LV-1035	PI: Kc 2-4, Ti 2-4 min
LIC-2014 / 2018 / 2021	C-201 draw-pan levels	small	-	< 2	Tight	product FICs	PI: Kc 2-4, Ti 1-3 min (pans run dry fast)
LIC-1110 / 2030	E-113 / E-201 steam drum	-	-	-	3-element	LV BFW	Level + steam flow FF + BFW flow cascade
LIC-1098	E-116 condensate pot	small	-	-	Tight	LV-1098	PI: Kc 2-3, Ti 2 min
LIC-2028 / 2029	D-201 hotwell water / slop	-	-	-	Tight (barometric seal)	LV-2028/2029	PI: Kc 2, Ti 3 min

5. Alarm philosophy (ISA-18.2 / IEC 62682)

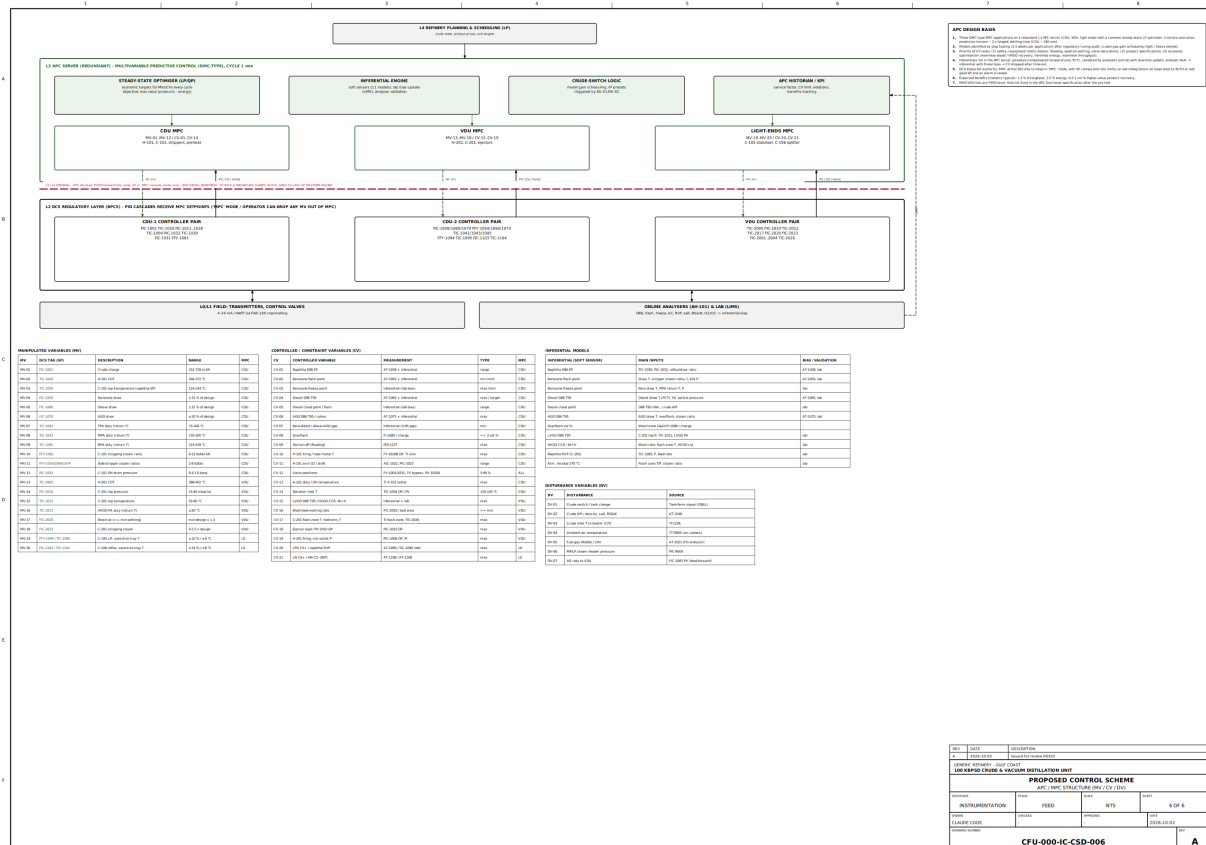
Alarm management follows the ISA-18.2 life cycle: philosophy -> identification -> rationalisation -> detailed design -> implementation -> operation -> maintenance -> monitoring & assessment -> MOC -> audit. Every alarm must indicate an abnormal condition requiring a timely operator action; status and information go to the journal, not the alarm list. Rationalisation is done in a workshop (process, operations, I&C) and recorded in the master alarm database (L3 alarm management server).

Priority	Consequence if no action	Time to respond	Target share	Annunciation
Emergency (1)	Safety / environment / major damage (pre-trip)	< 5 min	~ 5 %	Red, audible tone 1, cannot be shelved
High (2)	Off-spec, equipment damage, unit upset	5-15 min	~ 15 %	Orange, tone 2
Low (3)	Minor efficiency / quality loss	15-30 min	~ 80 %	Yellow, tone 3
Journal	Status, diagnostics, events	-	-	Event journal only

- **Performance targets (EEMUA 191):** average alarm rate ≤ 1 per 10 min per console in steady operation; peak ≤ 10 alarms in 10 min after an upset; $< 1\%$ of time in flood; no stale alarms > 24 h; no chattering (> 3 per minute).
- **Design rules:** deadbands and on/off delays per signal type (flow 2% , 5 s; level 2% , 10 s; pressure 1% , 5 s; temperature $1\text{ }^{\circ}\text{C}$, 15 s); SIS pre-trip alarms 5-10 % ahead of trip set point; deviation alarms for cascades only where the slave can saturate.
- **State-based alarming:** alarm sets switch with operating mode (start-up, normal, turndown, crude switch, shutdown) and with equipment status (spare pump, heater out of service).
- **Suppression and shelving:** designed suppression for consequential alarms after a trip (first-out kept); shelving time-limited (max 8 h) with shift-log review; emergency priority cannot be shelved.
- **SIS alarms:** SIS trips and first-out are displayed via the read-only SIS-DCS gateway; SIS bypasses and transmitter deviations alarm in the CCR.
- **KPIs:** monthly report from the alarm management server (top-10 bad actors, standing, chattering, flood periods), reviewed by operations.

6. Advanced process control (APC / MPC)

Three DMC-type multivariable predictive controllers run on a redundant L3 APC server with a common steady-state LP / QP optimiser: CDU (H-101, C-101, strippers, preheat), VDU (H-201, C-201, ejectors) and light ends (C-105, C-106). Execution 1 min; models from plant step tests after a regulatory tuning audit; crude-type gain scheduling for light / heavy blends. The MPC writes setpoints only to DCS loops in 'MPC' cascade mode with SP clamps and rate limits; a watchdog sheds all loops to AUTO at the last SP on loss of communication.



MV	DCS tag (SP)	Description	Range	MPC
MV-15	TIC-2012	C-201 top temperature	55-90 °C	VDU
MV-16	TIC-2017	HVGO PA duty (return T)	±20 °C	VDU
MV-17	FIC-2020	Wash oil (>= min wetting)	min-design x 1.3	VDU
MV-18	FIC-2023	C-201 stripping steam	0-1.5 x design	VDU
MV-19	FFY-1094 / TIC-1095	C-105 L/F, sensitive tray T	±10 % / ±8 °C	LE
MV-20	FIC-1103 / TIC-1104	C-106 reflux, sensitive tray T	±15 % / ±8 °C	LE

6.2 Controlled and constraint variables

CV	Variable	Measurement	Type	MPC
CV-01	Naphtha D86 EP	AT-1038 + inferential	range	CDU
CV-02	Kerosene flash point	AT-1055 + inferential	min limit	CDU
CV-03	Kerosene freeze point	inferential (lab bias)	max limit	CDU
CV-04	Diesel D86 T95	AT-1065 + inferential	max / target	CDU
CV-05	Diesel cloud point / flash	inferential (lab bias)	range	CDU
CV-06	AGO D86 T95 / colour	AT-1075 + inferential	max	CDU
CV-07	Kero-diesel / diesel-AGO gap	inferential (5-95 gap)	min	CDU
CV-08	Overflash	FI-1080 / charge	>= 3 vol %	CDU
CV-09	Section dP (flooding)	PDI-1237	max	CDU
CV-10	H-101 firing / tube metal T	FY-1020B OP, TI skin	max	CDU
CV-11	H-101 arch O2 / draft	AIC-1022, PIC-1023	range	CDU
CV-12	Valve positions	FV-1001/1031, TV bypass, PV-1032A	5-90 %	ALL
CV-13	A-101 duty / OH temperature	TI A-101 outlet	max	CDU
CV-14	Desalter inlet T	TIC-1004 OP / PV	125-145 °C	CDU
CV-15	LVGO D86 T95 / HVGO CCR, Ni+V	inferential + lab	max	VDU
CV-16	Wash-bed wetting rate	FIC-2020 / bed area	>= min	VDU
CV-17	C-201 flash zone T / bottoms T	TI flash zone, TIC-2026	max	VDU
CV-18	Ejector load / PV-2010 OP	PIC-2010 OP	max	VDU
CV-19	H-201 firing, coil outlet P	PIC-2006 OP, PI	max	VDU
CV-20	LPG C5+ / naphtha RVP	AT-1099 / TIC-1095 infer.	max	LE
CV-21	LN C6+ / HN C5- (IBP)	AT-1108 / AT-1109	max	LE

6.3 Disturbance variables

DV	Disturbance	Source
DV-01	Crude switch / tank change	Tank-farm signal (OSBL)
DV-02	Crude API / density, salt, BS&W	AT-1048
DV-03	Crude inlet T to heater (CIT)	TI-1226
DV-04	Ambient air temperature	TT-9005 (air coolers)
DV-05	Fuel-gas Wobbe / LHV	AT-1021 (FG analyser)
DV-06	MP/LP steam header pressure	PIC-9004
DV-07	AR rate to VDU	FIC-1083 PV (feed-forward)

6.4 Inferential models (soft sensors)

Inferentials are linear / PLS models on pressure-compensated temperatures (PCT), flows and heat-balance terms, executed in the APC server and validated by online analysers (fast bias) and laboratory results (slow bias via LIMS). On analyser failure the inferential continues with frozen bias; after a time-out the CV is dropped from the MPC.

Inferential	Main inputs	Bias / validation
Naphtha D86 EP	TIC-1030, PIC-1032, reflux/draw ratio	AT-1038, lab
Kerosene flash point	Draw T, stripper steam ratio, C-101 P	AT-1055, lab
Kerosene freeze point	Kero draw T, MPA return T, P	lab
Diesel D86 T95	Diesel draw T (PCT), HC partial pressure	AT-1065, lab
Diesel cloud point	D86 T95 infer., crude API	lab
AGO D86 T95	AGO draw T, overflash, steam ratio	AT-1075, lab
Overflash vol %	Wash-zone liquid FI-1080 / charge	-
LVGO D86 T95	C-201 top P, TIC-2012, LVGO PA	lab
HVGO CCR / Ni+V	Wash rate, flash zone T, HVGO cut	lab
Naphtha RVP (C-105)	TIC-1095, P, feed rate	lab
Atm. residue 370 °C-	Flash zone T/P, steam ratio	lab

6.5 Online analysers

Analysers are housed in the analyser house AH-101 (UPS-fed, HVAC, gas detection) close to the sample points; fast-loop sample systems with return to process; validation by automatic line sample and lab correlation.

Tag	Service	Type	Cycle	Use
AT-1022 / AT-2008	H-101 / H-201 flue gas O ₂ + CO	In-situ zirconia O ₂ + TDL CO (arch)	continuous	O ₂ trim, CO override, CV
AT-1037	D-102 boot sour water pH	pH probe (retractable)	continuous	Neutraliser trim
AT-1038	Unstab. naphtha D86 EP / RVP	Online distillation (ASTM D86 equiv.), AH-101	10-15 min	MPC CV-01, inferential bias
AT-1046	Desalted crude salt content	Online salt-in-crude (conductometric)	15 min	Desalter optimisation
AT-1047	Desalted crude BS&W	Microwave water-cut	continuous	Desalter performance
AT-1048	Charge crude API / salt / BS&W	Densitometer + salt + water-cut	continuous	Crude-switch DV (MPC)
AT-1049	Desalter brine oil-in-water	UV fluorescence	continuous	Brine quality / ETP
AT-1055	Kerosene flash / freeze point	Online flash (Pensky-Martens eq.) + freeze analyser	10-20 min	MPC CV-02/03
AT-1065	Diesel D86 T95 / cloud point	Online distillation + cloud point	15 min	MPC CV-04/05
AT-1075	AGO D86 T95 / colour	Online distillation + colorimeter	15 min	MPC CV-06
AT-1099	LPG C5+	Process GC	5 min	C-105 MPC CV
AT-1108 / AT-1109	LN C6+ & RVP / HN IBP & C5-	Process GC (shared, stream switching)	6 min	C-106 MPC CVs
AT-1021	Fuel gas Wobbe index / LHV	Fast calorimeter	< 30 s	Combustion FF, DV-05
AT-2032	Vacuum off-gas H ₂ S	UV / lead-acetate tape analyser	2 min	H-201 firing / SO ₂

7. Start-up, shutdown, crude switch and turndown

7.1 Start-up

- Cold circulation (crude through preheat, H-101 coils and C-101 bottoms back to slop) on flow control with level loops in AUTO; heater passes on FIC with SIF-101 armed (permissive: all pass flows > LL + margin).
- H-101 / H-201 light-off by BMS sequence: purge (5 volume changes, air flow $\geq 25\%$), pilot ignition with proof, main burners one by one; FG pressure on PIC-1021 in AUTO at minimum fire, COT ramp limited to $50\text{ }^{\circ}\text{C/h}$ by TIC-1020 SP ramp.
- Column pressure on PIC-1032 using FG make-up (PV-1032B); reflux established on FIC-1031 manual flow, TIC-1030 into cascade when the top temperature is meaningful; PAs started top to bottom on FIC with TICs in manual until duties stabilise.
- Side draws opened on FIC at low rate once trays are wet; stripping steam on ratio after product flow established; VDU started when AR quality is stable (vacuum pulled on ejectors with steam, PIC-2010 into AUTO; wash oil FIC-2020 established before H-201 COT exceeds $370\text{ }^{\circ}\text{C}$).
- MPC applications switched on after > 4 h of stable regulatory operation.

7.2 Normal shutdown

Reverse sequence with ramped SPs: MPC off, COT reduced to $300\text{ }^{\circ}\text{C}$ ($50\text{ }^{\circ}\text{C/h}$), draws closed progressively, heaters to minimum fire then burners out by BMS, circulation maintained until heater outlet < $200\text{ }^{\circ}\text{C}$, coil steam-out. Emergency shutdown is by SIS (ESD-1, SIF-901) per the C&E matrix.

7.3 Crude switch strategy

- Tank change signal (DV-01) and charge density AT-1048 (DV-02) trigger the crude-switch logic in the APC server: model gain set selected for the new crude, cut-point SPs pre-moved by the feed-forward, desalter chemical / wash ratios preset.
- COT and side-draw rates are moved with feed-forward on the measured density change; inferential bias updates are frozen for 2 h (analysers and lab confirm the new steady state).
- Regulatory layer: blending of tanks over ~30 min (OSBL) is preferred to a step change; FIC-1001 rate cut of 5-10 % during the switch is an operator option.

7.4 Turndown to 50 %

- H-101 / H-201 pass flow LL trips are set at 40 % of design pass flow, below the 50 % turndown, so the heaters can operate at turndown with margin; burners operate at ~ 45-50 % fire (turndown 3:1 burners).
- Control valves at 0.5 x normal flow: travel 25-67 % (all $\geq 10\%$, section 12).
- Orifice flow meters: at 50 % flow the dP is 25 % of span (rangeability 3:1 acceptable); pass flow and SIS flow transmitters are ranged so that the LL trip is above 20 % of span.
- Column internals: tray weeping limits the vapour rate to ~ 50-60 % of design; at turndown the stripping steam and pumparound ratios are kept, and the MPC keeps overflash and internal reflux minima.
- Pumps: minimum continuous flow by recirculation (P&ID); pumparound flows are kept above 60 % of design to keep exchangers in turbulent flow.

8. Safety instrumented systems (summary)

The SIFs below are implemented in the SIS (SIL 3-capable logic solver, separate from the DCS). Heater BMS functions run as separate applications on SIS hardware. SIL determination by LOPA is in CFU-000-IC-RPT-002; the cause & effect matrix is CFU-000-IC-CE-001.

SIF	Function	Initiators	Final elements	SIL (SRS)	LOPA RRF	LOPA SIL
SIF-101	H-101 low-low pass flow (any pass) -> trip fuel	FT-1011..1018 (2oo3 each pass)	XV-1021/1022 FG SSOV, XV-1026 pilot	SIL 2	250	SIL 2
SIF-102	H-101 low-low fuel gas pressure -> trip fuel	PT-1027A/B/C 2oo3	XV-1021/1022	SIL 1	25	SIL 1
SIF-103	H-101 high-high fuel gas pressure -> trip fuel	PT-1027A/B/C 2oo3	XV-1021/1022	SIL 1	25	SIL 1
SIF-104	H-101 loss of flame (all burners) -> trip fuel	BS-1028 (scanners)	XV-1021/1022/1026	SIL 2	125	SIL 2
SIF-105	H-101 high-high arch pressure / loss of ID fan -> trip	PT-1029A/B/C	XV-1021/1022	SIL 1	20	SIL 1
SIF-106	H-101 high-high COT -> trip fuel	TT-1020A/B/C 2oo3	XV-1021/1022	SIL 1	50	SIL 1
SIF-107	D-101A/B low-low interface (grid short) -> trip transformers	LT-1007B / LT-1008B	Desalter transformer CB	SIL 1	30	SIL 1
SIF-108	C-101 high-high bottom level -> stop P-101 / close FV-1001	LT-1082B	XV-1001 + P-101 trip	SIL 1	50	SIL 1
SIF-109	D-105 low-low level -> close LPG product XV (gas blow-by to LPG treating)	LT-1092B	XV-1093	SIL 1	30	SIL 1
SIF-110	C-105 high-high pressure -> cut reboiler steam	PT-1091B	XV-1096	SIL 1	20	SIL 1
SIF-201	H-201 low-low pass flow -> trip fuel	FT-2001..2004	XV-2006A/B	SIL 2	250	SIL 2
SIF-202	H-201 low-low FG pressure / flame failure -> trip	PT-2009A/B/C, BS-2008	XV-2006A/B	SIL 1	25	SIL 1
SIF-203	C-201 high-high bottom level -> trip FV-1083 / P-112	LT-2024B	XV-1083	SIL 1	50	SIL 1
SIF-204	Pump P-112/P-204 hot-pump emergency isolation (fire)	Manual HS / fire detection	EIV-1121/EIV-2041 (ROSOV)	SIL 1	50	SIL 1
SIF-901	Unit emergency shutdown (ESD-1, manual)	HS-9000 (CCR + field)	All unit trip valves, heater trips	SIL 2	-	SIL 2 (company standard, manual ESD - LOPA n/a)



The ICS follows the Purdue / ISA-95 model with a process DMZ (CFU-000-IC-BLK-001). The DCS has three redundant controller pairs in the field auxiliary room FAR-100 (CDU-1: charge, desalting, preheat, H-101, fuel gas; CDU-2: C-101, strippers, overhead, C-105, C-106; VDU: H-201, C-201, ejectors). The SIS logic solver (SIL 3 capable), the BMS applications and the F&G system are physically separate. Marshalling and system cabinets are in FAR-100; a redundant single-mode fibre ring connects FAR-100 to the OSBL central control room (CCR), where operator consoles, engineering stations and L3 servers are located.

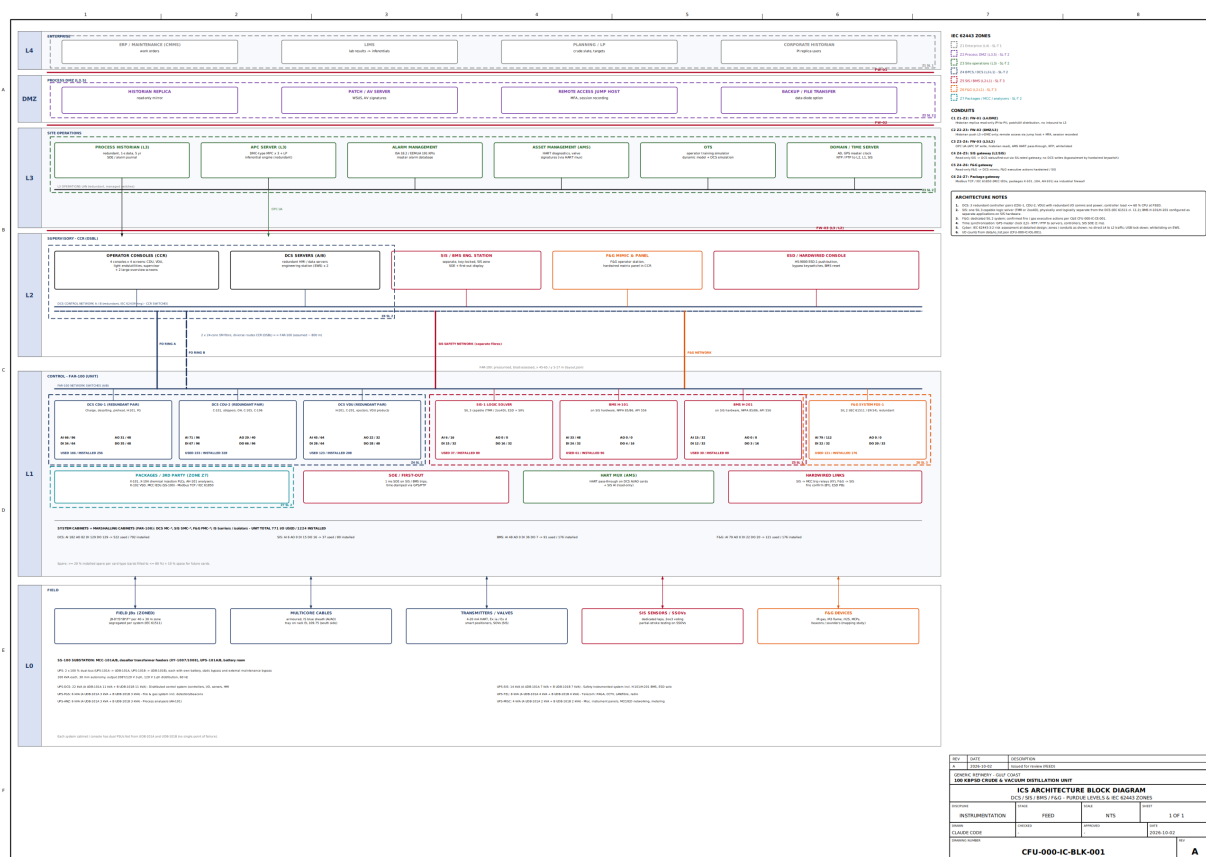


Figure 8 - ICS architecture (CFU-000-IC-BLK-001)

9.1 I/O summary (CFU-000-IC-IOL-001)

Controller	System	AI	AO	DI	DO	Used	Installed	Spare
CDU-1	DCS	66	31	34	35	166	256	54 %
CDU-2	DCS	71	29	67	66	233	328	41 %
VDU	DCS	45	22	28	28	123	208	69 %
SIS-1	SIS	6	0	15	16	37	80	116 %
BMS-H101	BMS	33	0	24	4	61	96	57 %
BMS-H201	BMS	15	0	12	3	30	80	167 %
FGS-1	F&G	79	0	22	20	121	176	45 %
TOTAL						771	1224	59 %

Points by source: instrument index: 373; I&C addition (add to index): 17; equipment.json (MCC hardwired interface): 256; C&E CFU-000-IC-CE-001: 8; F&G allowance (mapping study): 117.

9.2 IEC 62443 zones and conduits

Zone	Name	Target security level
Z1	Enterprise (L4)	SL 1
Z2	Process DMZ (L3.5)	SL 2
Z3	Site operations (L3)	SL 2
Z4	BPCS / DCS (L2-L1)	SL 2
Z5	SIS / BMS (L2-L1)	SL 3
Z6	F&G (L2-L1)	SL 3
Z7	Packages / MCC / analysers	SL 2

Conduit	Zones	Device	Permitted traffic / controls
C1	Z1-Z2	FW-01 (L4/DMZ)	Historian replica read-only (PI-to-PI), patch/AV distribution, no inbound to L3
C2	Z2-Z3	FW-02 (DMZ/L3)	Historian push L3->DMZ only; remote access via jump host + MFA, session recorded
C3	Z3-Z4	FW-03 (L3/L2)	OPC UA (APC SP write, historian read), AMS HART pass-through, NTP; whitelisted
C4	Z4-Z5	SIS gateway (L2/SIS)	Read-only SIS -> DCS status/first-out via SIL-rated gateway; no DCS writes (bypass/reset by hardwired keyswitch)
C5	Z4-Z6	F&G gateway	Read-only F&G -> DCS mimic; F&G executive actions hardwired / SIS
C6	Z4-Z7	Package gateway	Modbus TCP / IEC 61850 (MCC IEDs, packages X-101..104, AH-101) via industrial firewall

Power: all ICS cabinets have dual PSUs fed from UPS-101A / UPS-101B (data/electrical.json, 30 min autonomy); time synchronisation from a GPS master clock (NTP / PTP), 1 ms SOE in the SIS.

10. Instrument design criteria

- **Signals** - 4-20 mA with HART 7 for all analogue I/O (FOUNDATION Fieldbus H1 is an alternative for the DCS-only loops, not selected for FEED: HART gives one I/O per signal, simpler segregation and full asset management via HART multiplexers). Discrete: 24 VDC, NAMUR for IS proximity switches.
- **Hazardous area** - unit classified Zone 1 / 2, IIA / IIB T3 (HAC deliverable); transmitters Ex ia (intrinsically safe, galvanic isolators in FAR-100), solenoids and limit switches Ex ia or Ex d; analyser house AH-101 pressurised.
- **Pressure / dP** - smart electronic transmitters, 316 SS wetted parts (Hastelloy C for sour / H₂S service per NACE MR0103), remote seals with capillary for viscous / hot service (> 300 °C AR, VR, HVGO) and for vacuum service (absolute pressure transmitters, 0-100 mbar(a) for C-201).
- **Flow** - orifice plates (ISO 5167) with dP transmitters as default; Coriolis for chemical injection and wash water; wedge meters or venturi for viscous VR / AR / quench; vortex for steam; ultrasonic clamp-on not used for control. Pass flows: one orifice per pass with 1 BPCS + 3 SIS transmitters on separate taps.
- **Level** - guided-wave radar for drums and columns (two technologies for SIS: GWR / dP), displacer not used; desalter interface by GWR + multi-point density profiler; vacuum column boot by dP with remote seals and purge.
- **Temperature** - type K thermocouples (duplex) with head-mounted transmitters; RTD Pt100 for < 300 °C quality-critical points (sensitive trays, PCT inputs); heater tube-skin thermocouples (knife-edge) on each pass; flanged thermowells (ASME PTC 19.3 TW wake-frequency check).
- **Control valves** - globe, cage-guided, equal-percentage trim (linear for level / dP service where the valve dP is near-constant), class IV seat leakage (class V for tight shut-off duties), smart HART positioners with partial-stroke capability where also used as SIS final element (not adopted: SIS uses dedicated XVs). Noise <= 85 dBA at 1 m.
- **On-off / SIS valves** - fire-safe ball or gate valves, spring-return actuators, 24 VDC de-energise-to-trip SOVs, ZSO / ZSC limit switches; ROSOVs (EIV-1121 / 2041) fire-proofed for 30 min.
- **Analysers** - see section 6.5; sample systems with heated / insulated lines, fast loop and return.
- **Instrument air** - 7 barg, dew point -40 °C (basis); each valve with filter-regulator; air failure positions stated on P&IDs (FO / FC).

11. Control valve sizing (ISA-75.01.01)

Key control valves are sized for the maximum case (1.2 x normal flow) and checked at turndown (0.5 x normal). Liquid $C_v = Q / N_1 \times \sqrt{G / dP}$, with dP limited to the choked value $FL^2 (P_1 - FF P_v)$; gas / steam valves with the expansion factor $Y = 1 - x / (3 F_k xT)$, x limited to $F_k xT$. Valve dP allocations are FEED engineering judgement (stated); flows and densities from the H&MB (data/streams.json) and pump data (data/equipment.json). Body size: smallest size with travel <= 90 % at max flow (equal %, R = 50) and >= 10 % at turndown. Full calculation: CFU-000-IC-CAL-001 (xlsx).

Tag	Service	Normal flow	rho kg/m3	P1 bar(a)	dP bar	Cv norm	Cv max	Regime	Size	Rated Cv	Travel max / min %	Char.
FV-1001	Crude charge (P-101 disch.)	671.4 m3/h	846	25.0	3.0	412.1	494.6	normal	8" globe	750	89 / 67	Equal %
FV-1003	Desalter wash water (P-114)	35.1 m3/h	943	17.0	3.0	22.7	27.3	normal	2" globe	55	82 / 60	Equal %
LV-1007	D-101A brine to E-118	35.5 m3/h	950	12.51	6.51	15.7	18.8	normal	1.5" globe	32	59 / 25	Linear
FV-1031	C-101 reflux (P-103)	145.2 m3/h	698	8.7	2.0	99.1	119.0	normal	4" globe	200	87 / 64	Equal %

Tag	Service	Normal flow	rho kg/m3	P1 bar(a)	dP bar	Cv norm	Cv max	Regime	Size	Rated Cv	Travel max / min %	Char.
FV-1034	Unstab. naphtha to C-105 (P-104)	164.7 m3/h	698	17.7	3.0	91.8	110.2	normal	4" globe	200	85 / 62	Equal %
FV-1040	TPA circulation (P-106)	270.2 m3/h	617	10.2	1.5	200.4	240.4	normal	6" globe	430	85 / 63	Equal %
FV-1042	MPA circulation (P-107)	352.5 m3/h	591	10.2	1.5	255.8	307.0	normal	8" globe	750	77 / 55	Equal %
FV-1044	BPA circulation (P-108)	289.4 m3/h	573	10.2	1.5	206.7	248.1	normal	6" globe	430	86 / 64	Equal %
FV-1052	Kerosene product (P-109)	88.9 m3/h	772	10.0	2.0	63.8	76.6	normal	3" globe	120	89 / 66	Equal %
FV-1062	Diesel product (P-110)	101.0 m3/h	808	10.0	2.0	74.2	89.1	normal	4" globe	200	79 / 57	Equal %
FV-1083	Atm. residue to H-201 (P-112)	364.9 m3/h	704	20.0	2.5	223.9	268.6	normal	6" globe	430	88 / 66	Equal %
FV-2016	HVGO PA circulation (P-202)	237.3 m3/h	680	10.01	1.5	184.7	221.7	normal	6" globe	430	83 / 61	Equal %
FV-2025	Vacuum residue (P-204)	138.2 m3/h	784	14.01	3.0	81.6	98.0	normal	4" globe	200	82 / 59	Equal %
PV-1021	H-101 fuel gas to burners	5164 kg/h	3.57	4.5	1.49	99.2	119.1	Y=0.83	4" globe	200	87 / 64	Equal %
PV-1032 A	D-102 off-gas to FG / flare	11461 kg/h	3.21	1.7	0.49	404.6	485.5	Y=0.83	8" globe	750	89 / 67	Equal %
PV-1032 B	FG make-up to D-102	500 kg/h	3.57	4.5	2.8	8.6	10.3	Y=0.67	1.5" globe	32	71 / 49	Equal %
PV-2010	C-201 vacuum: NCG spill-back to J-201	506 kg/h	1.04	1.1	1.08	32.3	38.7	choked	3" globe	120	71 / 49	Equal %
FV-1096	HP steam to E-116 (C-105 reboiler)	22926 kg/h	13.64	42.41	6.36	97.7	117.2	Y=0.92	4" globe	200	86 / 64	Equal %

Notes: (1) no liquid valve is choked or flashing at the allocated dP (LV-1007 brine checked against Pv at 136 °C); (2) PV-2010 is choked (vacuum service) - noise / velocity check and possibly a 2-stage trim at detailed design; (3) PV-1032A / PV-1032B sizing cases are assumptions (no H&MB flow - normally closed); (4) viscosity correction (FR) to be checked for VR (FV-2025) with vendor data.

12. Typical loop diagrams

Three typical loop diagrams (A3, ISA-5.4) are issued to fix the wiring philosophy: CFU-100-IC-LD-001 (FIC-1031 reflux flow, FT + FV with smart positioner), CFU-100-IC-LD-002 (TIC-1020 / PIC-1021 COT cascade) and

CFU-100-IC-LD-003 (SIF-101 pass 1 low-low flow, 2003 FT -> SIS -> SSOVs with solenoids). JB, multicore, marshalling terminals and I/O card / channel come from the I/O list.

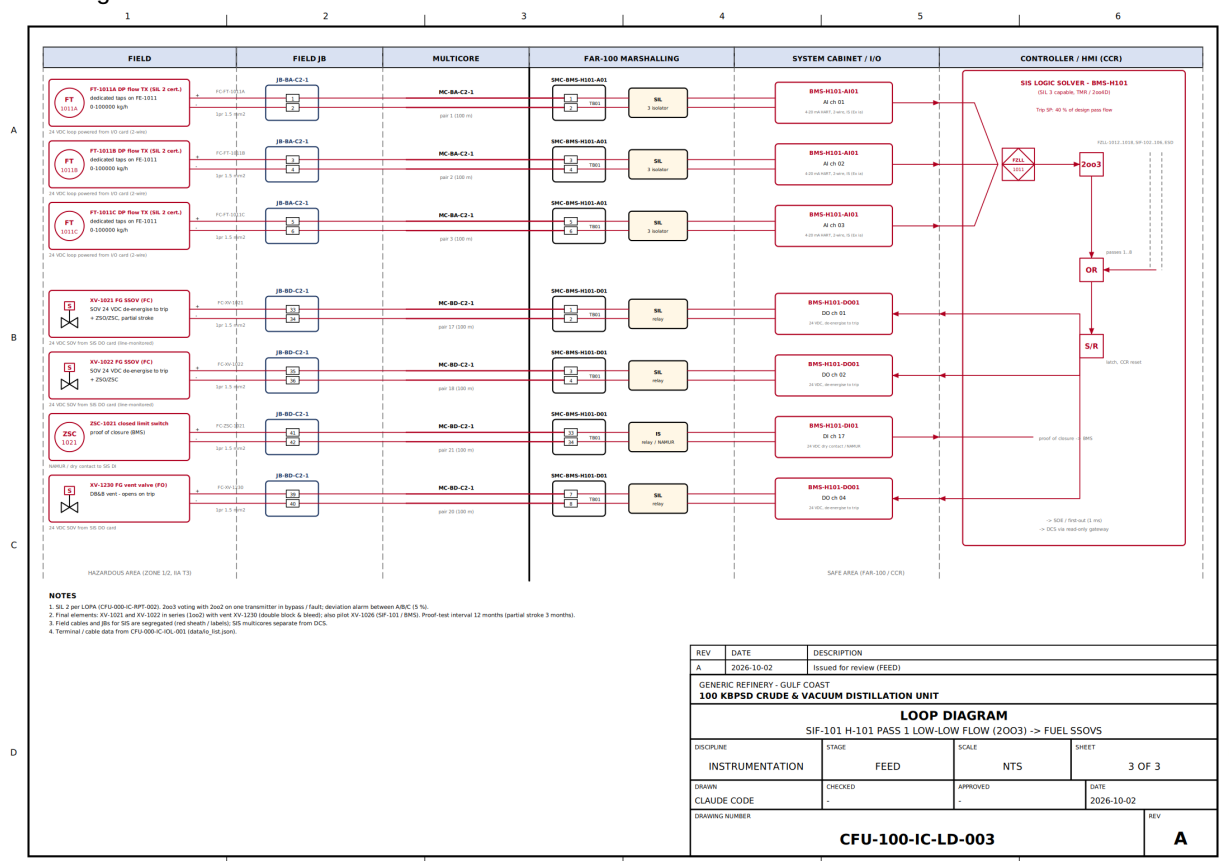


Figure 9 - Loop diagram SIF-101 (CFU-100-IC-LD-003)

13. Open items, holds and interface notes

- I&C additions to be added to the instrument index (CFU-000-IC-IDX-001) by the P&ID discipline: analysers AT-1021, AT-1038, AT-1046..1049, AT-1055, AT-1065, AT-1075, AT-1099, AT-1108, AT-1109, AT-2032; fuel gas flow FT-1021 / FT-2006 (cross-limiting); ambient TT-9005; analyser house alarm XA-9101; computing blocks FY-1011A..1018B, FY-1020A/B, FY-1021A/C, PY-1021A/B, FY-1025A..C, FFY-1054/1064/1074/1081/1094, FY-2020, PY-1032; heater snuffing steam HV-1290 / HV-2190.
- Instrument index: flame scanners BS-1028 / BS-2008 are listed with system 'DCS' - they are BMS (SIS logic solver) inputs; UZ-9000 (ESD logic) carries signal '4-20 mA HART (SIS AI)' - it is a logic function with no field I/O; ZSC-1231 / ZSC-1232 duplicate the limit switches already included with XV-1021 / XV-1022 (signal 'DO + 2 x DI') - counted once in the I/O list.
- F&G field devices are not yet in the instrument index (only GD-1284 / 1285 and BY-1121 / 2041); the I/O list carries a per-fire-zone allowance until the F&G mapping study is done.
- Motor interfaces (run / fault DI, start / stop DO) are hardwired allowances derived from equipment.json; a serial / IEC 61850 MCC interface would remove ~ 250 hardwired points (decision at detailed design).
- Burner FG pressure trip settings (0.15 / 2.2 barg), the burner curve and minimum fire are FEED assumptions pending burner vendor data.
- PV-1032A / PV-1032B and PV-2010 sizing flows are assumptions (no H&MB case for normally-closed duties).
- CCR distance (fibre route ~ 800 m) is an assumption; CCR building, consoles and L3 servers are OSBL scope shared with the refinery.
- Drawing frame: the common A3 title block (cfu/drawing/sheet.py, 170 x 50 mm) leaves the drawing-number row only 4 mm high so the number overlaps the frame; the I&C loop diagrams use a local subclass with the A1-size block. Fix in sheet.py recommended (process / drawing owner).

- LOPA frequencies, consequence categories and IPL credits are FEED judgements; to be confirmed in the SIL workshop after HAZOP close-out.

Appendix A - Principal control loop list

Tag	Service	Measured	Final element / output	Notes
FIC-1001	Crude charge flow (unit throughput)	P-101 discharge	FV-1001	Master; sets heater pass SPs via ratio
FFIC-1002	Demulsifier injection ratio to crude	X-101	X-101 stroke	Ratio to FIC-1001
FFIC-1003	Wash water ratio to crude (5 LV%)	P-114 discharge	FV-1003	Ratio to FIC-1001
TIC-1004	Desalter inlet temperature	D-101A inlet	TV-1004 (E-105 bypass)	Limits 125-145 C
PDIC-1005	1st-stage mix valve dP	Mix valve D-101A	PDV-1005	0.5-1.5 bar
PDIC-1006	2nd-stage mix valve dP	Mix valve D-101B	PDV-1006	0.5-1.5 bar
LIC-1007	D-101A water/oil interface	D-101A	LV-1007 brine	Guided-wave radar + profiler
LIC-1008	D-101B water/oil interface	D-101B	LV-1008 to 1st-stage wash	
PIC-1009	Desalter pressure (above crude vapour pressure)	D-101B outlet	PV-1009 P-102 suction	Backpressure
FFIC-1010	Caustic injection ratio	X-102	X-102 stroke	Ratio to FIC-1001
FIC-1011	H-101 pass 1 flow	Pass 1 inlet	FV-1011	Pass balancing via TDIC-1019
FIC-1012	H-101 pass 2 flow	Pass 2 inlet	FV-1012	
FIC-1013	H-101 pass 3 flow	Pass 3 inlet	FV-1013	
FIC-1014	H-101 pass 4 flow	Pass 4 inlet	FV-1014	
FIC-1015	H-101 pass 5 flow	Pass 5 inlet	FV-1015	
FIC-1016	H-101 pass 6 flow	Pass 6 inlet	FV-1016	
FIC-1017	H-101 pass 7 flow	Pass 7 inlet	FV-1017	
FIC-1018	H-101 pass 8 flow	Pass 8 inlet	FV-1018	
TDIC-1019	H-101 pass outlet temperature balancing	Pass outlets TI-1011..1018	Biases FIC-1011..1018	Total flow held
TIC-1020	H-101 coil outlet temperature (COT)	Combined outlet	Cascade to PIC-1021	Feed-forward from FIC-1001
PIC-1021	H-101 fuel gas burner pressure	FG to burners	PV-1021	Slave of TIC-1020; low select vs min-fire
AIC-1022	H-101 flue gas O2 (excess-air trim)	Arch O2/CO analyser	FD fan inlet vanes	Cross-limited with fuel
PIC-1023	H-101 arch draft	Arch pressure	ID fan speed / stack damper	-2.5 mmH2O
TIC-1024	Stripping-steam superheat outlet	SS coil outlet	TV-1024 desuperheat	350 C
FIC-1025	H-101 combustion air flow	FD fan discharge	FD inlet vanes	Cross-limit with fuel
TIC-1030	C-101 top temperature (naphtha end point)	C-101 top	Cascade to FIC-1031	APC handle
FIC-1031	C-101 reflux	P-103 discharge	FV-1031	Slave of TIC-1030
PIC-1032	OH drum pressure (split range)	D-102	PV-1032A off-gas / PV-1032B FG make-up	Split range
LIC-1033	D-102 hydrocarbon level	D-102	Cascade to FIC-1034	Averaging
FIC-1034	Unstabilised naphtha to C-105	P-104 discharge	FV-1034	
LIC-1035	D-102 boot interface	D-102 boot	LV-1035 sour water	
FFIC-1036	OH neutraliser ratio / pH	X-103	X-103 stroke	pH AIC-1037 trim
AIC-1037	OH sour-water pH	D-102 boot	Trims FFIC-1036	pH 5.5-6.5
FIC-1040	TPA circulation	P-106 discharge	FV-1040	

Tag	Service	Measured	Final element / output	Notes
TIC-1041	TPA return temperature (duty)	TPA return	TV-1041 E-101 bypass	
FIC-1042	MPA circulation	P-107 discharge	FV-1042	
TIC-1043	MPA return temperature	MPA return	TV-1043 E-106 bypass	
FIC-1044	BPA circulation	P-108 discharge	FV-1044	
TIC-1045	BPA return temperature	BPA return	TV-1045 E-113 bypass	
FIC-1050	Kerosene draw to C-102	Draw line	FV-1050	Cut point; APC from TI-1053 / flash
LIC-1051	C-102 bottom level	C-102	Cascade to FIC-1052	
FIC-1052	Kerosene product	P-109 discharge	FV-1052	
FIC-1054	C-102 stripping steam	Steam to C-102	FV-1054	Ratio to product (6 lb/bbl)
FIC-1060	Diesel draw to C-103	Draw line	FV-1060	Cut point (T95)
LIC-1061	C-103 bottom level	C-103	Cascade to FIC-1062	
FIC-1062	Diesel product	P-110 discharge	FV-1062	
FIC-1064	C-103 stripping steam	Steam to C-103	FV-1064	Ratio
FIC-1070	AGO draw to C-104	Draw line	FV-1070	
LIC-1071	C-104 bottom level	C-104	Cascade to FIC-1072	
FIC-1072	AGO product	P-111 discharge	FV-1072	
FIC-1074	C-104 stripping steam	Steam to C-104	FV-1074	Ratio
FI-1080	Overflash (wash-zone liquid) flow	Overflash pan	-	Min overflash alarm; APC constraint
FIC-1081	C-101 bottom stripping steam	Steam to C-101	FV-1081	Ratio to AR (10 lb/bbl)
LIC-1082	C-101 bottom level	C-101 sump	Cascade to FIC-1083	
FIC-1083	Atm. residue to H-201	P-112 discharge	FV-1083	Ratio-splits H-201 passes
FIC-1090	Stabiliser feed	P-104 discharge (via E-114)	FV-1034 (shared)	Same as FIC-1034
PIC-1091	C-105 pressure (hot-vapour bypass)	D-105	PV-1091	
LIC-1092	D-105 level	D-105	Cascade to FIC-1093 LPG	
FIC-1093	LPG product	P-115 discharge	FV-1093	
FIC-1094	C-105 reflux	P-115 discharge	FV-1094	Ratio to feed
TIC-1095	C-105 sensitive tray temperature (RVP)	Tray 15	Cascade to FIC-1096	
FIC-1096	HP steam to E-116	Steam to E-116	FV-1096	Condensate pot LIC-1098
LIC-1097	C-105 bottom level	C-105	LV-1097 feed to C-106	
LIC-1098	E-116 condensate pot level	Pot	LV-1098	
PIC-1100	C-106 / D-106 pressure	D-106	PV-1100 (flooded condenser)	
LIC-1101	D-106 level	D-106	Cascade to FIC-1102 LN	
FIC-1102	Light naphtha product	P-116 discharge	FV-1102	
FIC-1103	C-106 reflux	P-116 discharge	FV-1103	
TIC-1104	C-106 sensitive tray temperature	Tray 30	Cascade to FIC-1105	
FIC-1105	MP steam to E-117	Steam to E-117	FV-1105	
LIC-1106	C-106 bottom level	C-106	Cascade to FIC-1107 HN	
FIC-1107	Heavy naphtha product	P-117 discharge	FV-1107	
LIC-1110	E-113 steam-generator level	E-113	LV-1110 BFW	3-element

Tag	Service	Measured	Final element / output	Notes
PIC-1111	E-113 MP steam pressure	E-113 outlet	PV-1111	
FIC-2001	H-201 pass 1 flow	Pass 1 inlet	FV-2001	
FIC-2002	H-201 pass 2 flow	Pass 2 inlet	FV-2002	
FIC-2003	H-201 pass 3 flow	Pass 3 inlet	FV-2003	
FIC-2004	H-201 pass 4 flow	Pass 4 inlet	FV-2004	
TIC-2005	H-201 COT	Combined outlet	Cascade to PIC-2006	
PIC-2006	H-201 fuel gas pressure	FG to burners	PV-2006	
FIC-2007	H-201 coil (velocity) steam	Steam to passes	FV-2007	Per pass FI
AIC-2008	H-201 flue gas O2	Arch O2	Stack damper	
PIC-2010	C-201 top pressure (vacuum)	C-201 top	PV-2010 off-gas recycle to J-201 suction	
FIC-2011	LVGO PA circulation	P-201 discharge	FV-2011	
TIC-2012	C-201 top temperature	C-201 top	Cascade to TIC-2013	
TIC-2013	LVGO PA return temperature	LVGO PA return	A-201 fan pitch / bypass	
LIC-2014	LVGO draw-pan level	C-201 pan	Cascade to FIC-2015	
FIC-2015	LVGO product	P-201 discharge	FV-2015	
FIC-2016	HVGO PA circulation	P-202 discharge	FV-2016	
TIC-2017	HVGO PA return temperature	HVGO PA return	TV-2017 E-108 bypass	
LIC-2018	HVGO draw-pan level	C-201 pan	Cascade to FIC-2019	
FIC-2019	HVGO product	P-202 discharge	FV-2019	
FIC-2020	Wash oil to wash bed (critical, min-flow)	Wash spray header	FV-2020	Low-flow alarm & APC constraint
LIC-2021	Slop-wax pan level	C-201 pan	Cascade to FIC-2022	
FIC-2022	Slop wax product	P-203 discharge	FV-2022	
FIC-2023	C-201 bottom stripping steam	Steam to C-201	FV-2023	Ratio
LIC-2024	C-201 bottom level	C-201 boot	Cascade to FIC-2025	
FIC-2025	Vacuum residue product	P-204 discharge	FV-2025	
TIC-2026	C-201 bottoms temperature (coking control)	C-201 boot	Cascade to FIC-2027	Max 365 C
FIC-2027	VR quench flow	Quench return	FV-2027	
LIC-2028	D-201 hotwell water level	D-201	LV-2028 P-205	
LIC-2029	D-201 slop oil level	D-201	LV-2029 P-206	
LIC-2030	E-201 steam-generator level	E-201	LV-2030 BFW	
PIC-2031	Vacuum off-gas to H-201	D-202	PV-2031	
PIC-9001	Fuel gas header pressure	D-103	PV-9001	
LIC-9002	D-103 liquid level	D-103	LV-9002	
LIC-9003	D-104 flare KO drum level	D-104	Pump-out on/off	
PIC-9004	LP steam header pressure (unit)	LP header	PV-9004 let-down	